

Fat Tails and Power Laws in Financial Returns

An easy-to-follow guide, and the link to the Student-t distribution

Power law · Tail index α · Higher moments · Student-t

Key question: why can relying only on a normal distribution lead us to underestimate financial risk? Because the probability of very large gains or losses does not disappear as quickly as the normal model suggests.

How to Read This Note

Five steps, from the idea of a fat tail to the Student-t distribution

Power Law → Fat Tail → Extreme Events Matter

- ① What a fat tail is
- ② Why a power law means probability declines slowly
- ③ What a tail index of about 3–4 means
- ④ Why higher moments can become unstable
- ⑤ How the Student-t distribution represents power-law tails

Three Statements, One Story

The three lines in the original box are not separate facts



Because the probability of large events declines slowly, extreme risk deserves separate attention.

Financial markets may look calm most of the time, yet storm-scale moves occur more often than a normal model would suggest. Looking only at the mean and the standard deviation can therefore miss an important part of risk.

The Original Wording, in Plain Language

What each line on the slide actually claims

Original wording	Plain language
“The tail follows a power law, not an exponential.”	The probability of a very large event does not fall to zero quickly. It declines as a power of event size.
“A tail index around three to four is typical.”	Tails are fairly heavy. Values of $\alpha \approx 3-4$ are often reported for financial returns.
“The mean exists; higher moments barely do.”	Mean and variance may exist, but skewness and kurtosis can be unstable or infinite.

What Is a Power Law?

Probability declines as a power of event size

$$P(|R| > x) \propto 1 / x^\alpha$$

α (alpha) is the tail index. As x grows, the probability of a large event falls.

Reading the formula

- x is the size of the move, measured in the usual units of return.
- The left side is the probability of a move larger than x , in either direction.
- α controls how fast that probability shrinks.

Why call it a “power” law?

- The size enters as a power: x^3 , x^4 , $x^{2.5}$.
- With $\alpha = 3$ the rule reads $P(|R| > x) \propto 1 / x^3$.
- The exponent, not a constant rate, sets the speed of decay.

The Numbers Make It Easier

Relative probability under a power law with $\alpha = 3$

Event size	Power law: $1 / x^3$	Meaning
1×	1	Baseline
2×	1/8	One eighth of the baseline
3×	1/27	One twenty-seventh
5×	1/125	One hundred and twenty-fifth
10×	1/1000	One thousandth — small, but not zero

Large events certainly become rarer. But under a power law the probability declines slowly: an event ten times larger still keeps a small, non-zero share of the probability.

How Fast Does the Probability Vanish?

Same starting point, very different speeds of decay

$P(|R| > x)$, log scale

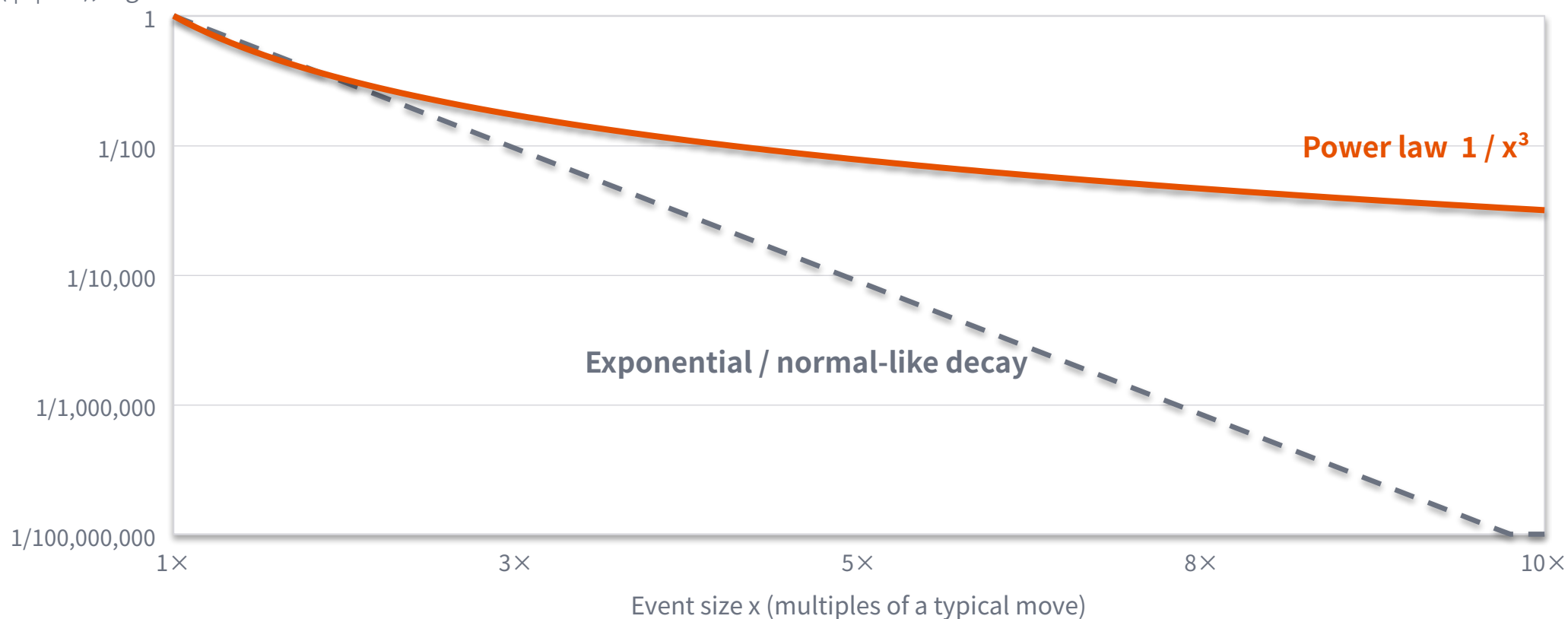


Figure 1. Under a power law the probability declines much more slowly for large x . The vertical axis is on a log scale.

How Is This Different from a Normal Distribution?

Both make large events rarer — the difference is how quickly

Normal distribution

- The probability of a large move declines very quickly.
- Beyond a few standard deviations it is effectively zero.

Power law

- The probability declines slowly.
- Extreme moves keep a small but real share of the probability.

Analogy · human height

- Most observations cluster around the average.
- People 3 m or 5 m tall essentially do not exist.
- This is normal-distribution intuition.

Analogy · financial markets

- Moves of 1–2% are the everyday case.
- Moves of 8%, 10%, even 20% do occur.
- This is fat-tail intuition.

Same Centre, Very Different Tails

Normal versus Student-t with $\nu = 3$, zoomed on the right tail

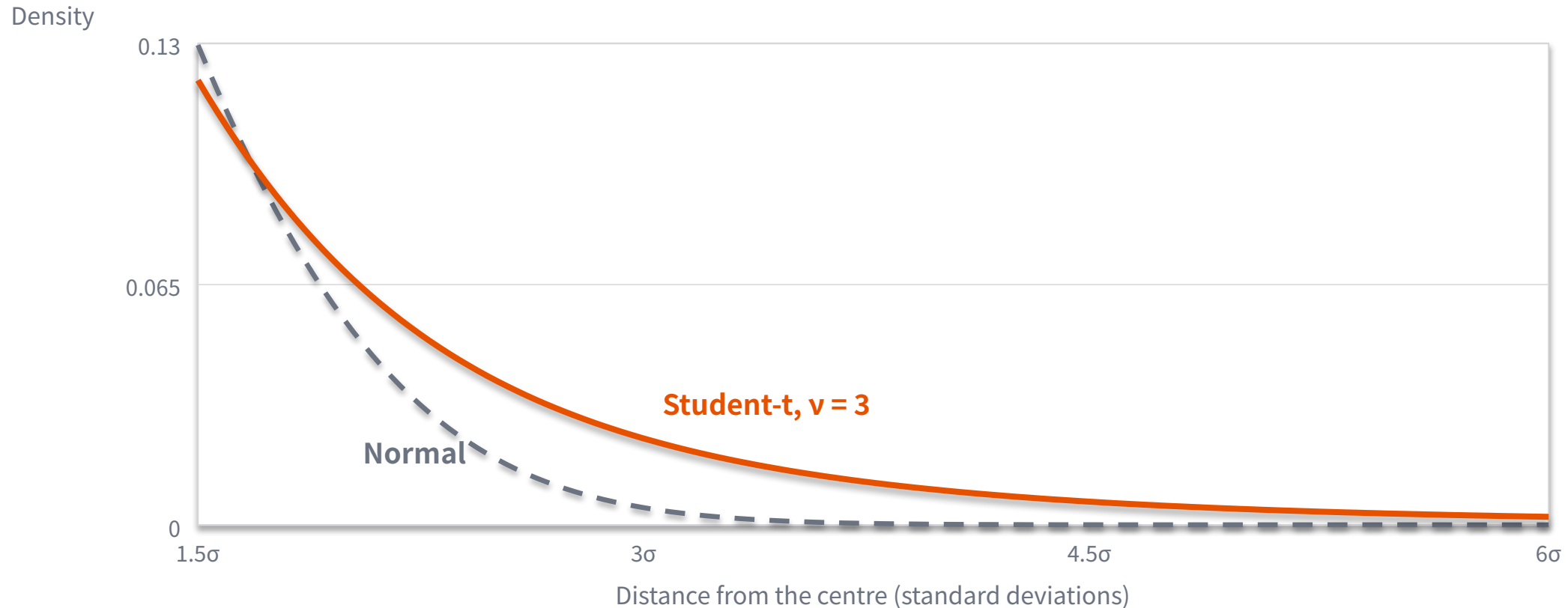


Figure 2. The Student-t may look similar near the centre, but it leaves much more probability in the tails.

What Is the Tail Index α ?

One number for how quickly the tail thins out

$$P(|R| > x) \propto 1 / x^\alpha$$

A smaller α means a slower decline, and therefore a heavier tail.

α	Tail thickness	Intuition
Smaller	Heavier	Large events keep relatively more probability
Larger	Thinner	The probability of large events declines faster

The smaller α is, the heavier the tail. “3–4” is an empirical range often discussed for financial heavy tails — not a universal constant. Estimates vary across assets, sample periods and data frequencies.

“The Mean Exists; Higher Moments Barely Do”

A moment is just one way of summarising a distribution with a number

Order	Statistic	Plain meaning
1st	Mean	The typical level
2nd	Variance	How much it fluctuates
3rd	Skewness	Whether it tilts to one side
4th	Kurtosis	How strongly extremes matter

$k < \alpha \rightarrow$ the k -th moment is finite

If $\alpha = 3.5$, the 1st, 2nd and 3rd moments can be finite while the 4th need not be.

Why Can Higher Moments Be Unstable?

One extreme observation can move the whole statistic

What happens

- Kurtosis computed from ordinary periods can look reasonably stable.
- Adding a single crisis-scale day can make it jump sharply.
- A statistic that depends on the fourth power is dominated by the largest observation.

Suggested revision to the slide

- The original line is highly compressed.
- Clearer: “Mean and variance may exist, but higher moments can be unstable or infinite.”
- This keeps the point without overstating it.

Practical reading: report the mean and variance as usual, but treat skewness and kurtosis as diagnostics whose values depend on whether a crisis sits inside the sample.

Can a Power Law Be Represented by a Student-t?

Answer: yes — it is the standard worked example

The Student-t is a standard distribution with power-law tails, so it is often better suited than the normal for representing fat-tailed financial returns.

$$P(|X| > x) \propto x^{-\nu} \quad (\text{for large } x)$$

ν (ν) is the degrees of freedom of the Student-t.

What this buys us

- The degrees of freedom ν play the role of the tail index α .
- Choosing ν is therefore a statement about how heavy the tails are.
- The distribution stays easy to estimate and to simulate.

Intuition for ν

One dial that runs from very heavy tails to the normal distribution

ν	Tail	Relation to the normal distribution
3	Very heavy	Much more probability remains in the extremes
4	Heavy	Still clearly fat-tailed
10	Much thinner	Close to the normal distribution
∞	Normal	The Student-t converges to the normal

$\nu \downarrow \rightarrow$ fatter tails $\nu \uparrow \rightarrow$ closer to normal

Student-t and the Existence of Moments

The degrees of freedom decide which moments exist at all

Condition	Meaning
$\nu > 1$	The mean exists
$\nu > 2$	The variance exists
$\nu > 3$	The third moment exists
$\nu > 4$	The fourth moment exists

An important distinction

- With $\nu = 3.5$ the mean and variance exist, but the fourth moment does not — so kurtosis need not be finite.
- The Student-t is one specific way of realising a power-law tail. Not every power-law distribution is a Student-t.

A Recommended Order for the Classroom

Build the intuition first, then attach the algebra

Steps 1–4

- ① Start with a realistic “ordinary day versus crash day” example.
- ② Ask the question: how quickly does the probability shrink?
- ③ Work the $1/x^3$ arithmetic: $2\times \rightarrow 1/8$, $3\times \rightarrow 1/27$.
- ④ Use the figure to show that the normal goes to zero far faster.

Steps 5–7

- ⑤ Attach the tail-index intuition: a smaller α means a heavier tail.
- ⑥ Close with the Student-t at $v = 3$ or 4 as a real model with power-law tails.
- ⑦ Treat higher moments as optional, centred on “one extreme day moves the kurtosis”.

Thirty-second version: “A power law says large events do become rarer, but their probability shrinks slowly. Financial markets show exactly this kind of fat tail, so crashes the normal model treats as near-impossible do happen from time to time. The Student-t is the standard distribution with power-law tails.”

One-Page Summary

Seven things worth carrying out of this note

Concept	What to remember
Power law	$P(R > x) \propto 1/x^\alpha$. The probability of a large event declines as a power of its size.
Fat tail	A tail that leaves more probability on extreme events than the normal distribution does.
Tail index α	The smaller it is, the heavier the tail. $\alpha \approx 3-4$ is a commonly cited empirical range.
Higher moments	When α is low, skewness and kurtosis can be unstable or infinite.
Student-t	The standard distribution with power-law tails; smaller v means fatter tails.
Normal versus t	The normal tail probability falls very fast; the Student-t tail falls slowly.
What it means for risk	Relying on a normal VaR can understate extreme risk. The tail deserves separate treatment.

TAKEAWAY

Risk lives in the tails.

The central risk in financial markets shows up not on the average day but on the rare, very large one. Power laws and the Student-t distribution describe that fact better than the normal distribution does.